Public Perception of the "Soul" in AI-Generated Art: An Empirical Study Based on Multiple Linear Regression

Abstract

With the advancement of generative-AI technologies in the artistic domain, the question of whether AI-created art possesses a "soul" has become a widely discussed public topic. This paper investigates public perceptions of the "soul" of AI art. Selected influencing factors include recognition of traditional artistic views, familiarity with AI technologies, frequency of emotional responses, acceptance of AI as an artistic tool, prior art-related exposure, and gender. Based on 59 valid online-survey responses, this study adopts reliability and validity tests, multiple-linear-regression analysis, and regression diagnostics to explore how these predictors shape public perception of artistic "soul". The results show that most respondents hold positive attitudes toward the soul-bearing capacity of AI art, with 61 % believing AI art may possess a soul. Recognition of traditional artistic views and gender exert statistically significant effects: higher endorsement of traditional artistic perspectives and male gender correlate with greater tendency to attribute a soul to AI art. In contrast, familiarity with AI technology, frequency of emotional responses, acceptance of AI as an artistic tool, and art-related background exhibit no significant predictive power. The overall multiple-regression model is statistically significant ($F=11.18$，$p<0.001$) with an adjusted R^2 of 0.513, indicating satisfactory explanatory power. These findings contribute to understanding public opinions on AI-generated art and offer references for future art outreach and art education.

Keywords: AIGC; AI-generated art; perception of artistic soul; multiple linear regression; traditional artistic views

1. Research Background and Research Questions

Research Background

As generative-AI techniques (e.g., GAN and Diffusion models) penetrate deeply into artistic creation, whether artificial intelligence can produce art endowed with a "soul" has emerged as a core debate among academia, art circles, and the general public. Diverse AI art tools are widely accessible. Ordinary audiences frequently encounter AIGC artworks, and some users even generate artistic outputs themselves. Accordingly, audiences are increasingly concerned whether AI-produced works

can convey human-like emotion — in other words, whether they possess a "soul".

Research Question

What factors shape public recognition and perception of the proposition that artificial intelligence can create soul-infused artworks?

2. Literature Review and Theoretical Hypotheses

2.1 Literature Review

A. Pure-theoretical Studies

Scholarly discourse over whether AI can create art with a soul falls into two major camps: the affirmative perspective and the negative perspective.

- **Negative perspective: the traditional-art-theory stance**

  This camp holds that the "soul" of art originates uniquely from human consciousness, emotion and lived experience. Since artificial intelligence lacks subjective intentionality and genuine life experience, AI-generated works cannot possess a soul in the traditional artistic sense. Scholars note that although viewers may have emotional reactions to AI art, such affective experiences tend to be weaker than those elicited by human-made art and are heavily contingent on contextual information (for instance, knowing whether a work is human-authored). This suggests fundamental differences in subjectivity and emotional expressiveness between human creators and AI systems.

  Creativity research also demonstrates that human creative activity is inherently intertwined with motivation, emotion and lived experience — dimensions that resist computational replication (Boden, 2004).

- **Affirmative perspective: technological empowerment and aesthetic construction**

  The opposing viewpoint argues that AI art can develop new forms of aesthetic value as technologies evolve. According to Floridi & Chiriatti (2020), if AI-generated artifacts evoke emotional engagement and meaning-making among audiences, such works can be regarded as artistically valid from a social-acceptance standpoint. Other scholars point out that machine-generative systems can yield unanticipated creative outputs and thereby expand the boundaries of conventional human creativity (Boden, 2016).

B. Review-oriented Studies

Review papers in AI-art research inform the selection of independent variables for the present investigation. Existing scholarship centers on technological creativity, public perceptions, affective experiences, and value judgments (Elgammal et al., 2017). AI literacy and technological comprehension have also grown into important research themes (Long & Magerko, 2020).

C. Descriptive Studies

Descriptive research applies questionnaires and case analyses to map public opinions on whether artworks contain a soul, documenting attitudes and group differences and furnishing empirical foundations for subsequent inquiry.

- **International literature**: Public reception of AI art constitutes a vital empirical-research domain. Studies reveal that audiences downgrade their aesthetic evaluations once informed that a work is AI-produced; aesthetic-emotional responses differ systematically between human-authored and AI-authored pieces. Compared with art professionals, lay audiences tend to exhibit greater tolerance and openness toward AI art (Millet et al., 2023).
- **Domestic literature**: Chinese-context studies indicate that contextual framing — such as explicit labeling of works as "AI-generated" — significantly modulates public aesthetic attitudes (Hong & Curran, 2019).

D. Explanatory Studies

Explanatory research investigates determinants of public perceptions of soul in AI art and tests underlying mechanisms; such literature directly guides variable selection in this paper. Four core sets of predictors are highlighted:

- **Traditional artistic conceptions**: Individuals who strongly subscribe to the view that art must spring from human subjectivity tend to deny the spiritual merit of AI art. Even when viewers feel emotions toward AI works, they may attribute those feelings to contextual framing rather than intrinsic qualities of the AI output.
- **Perceptions of AI technology**: Higher AI literacy may reduce technological unfamiliarity and improve public attitudes, yet it may simultaneously reinforce perceptions of AI as merely instrumental, yielding mixed effects (Long & Magerko, 2020).
- **Affective experience**: Experimental-psychology research establishes that aesthetic emotional experience constitutes a critical basis for judging an artwork as meaningful (Silvia, 2005; Pelowski et al., 2017).
- **Art-related literacy**: Audiences with professional art backgrounds tend to judge AI art more critically than general-public respondents (Millet et al., 2023).

2.2 Theoretical Dialogue

While existing literature offers diverse viewpoints, notable research gaps create opportunities for the present study:

- **Incomplete variable coverage**: Prior work emphasizes traditional-artistic beliefs, technological cognition, and affective experience but largely overlooks *acceptance of AI as an artistic tool*, a factor that may directly shape judgments about whether AI art has a soul.

- **Limited sample representativeness**: Most existing empirical research samples draw heavily on urban residents or art-industry practitioners.
- **Shallow mechanism analysis**: Many prior studies rely on univariate analyses or simple regression specifications rather than multivariate regression frameworks.
- **Disconnect between theory and empirics**: Empirical investigations often adopt a single theoretical lens (either traditional-art theory or techno-optimism), without integrating competing theoretical perspectives.

2.3 Theoretical Stance and Innovations

2.3.1 Theoretical Stance

This paper endorses the **constructivist view of artistic soul** proposed by Floridi & Chiriatti (2020): the soul of art is not an exclusive property of human creation but emerges interactively from *art-work content, human perception, and social context*. In other words, AI art can be perceived as soul-bearing insofar as it triggers human emotional resonance and conveys spiritual connotations. This perspective moves beyond essentialist frameworks (Demmer et al., 2023) that tie artistic soul exclusively to human consciousness, while avoiding the excessive idealism of techno-optimism (Boden, 2016).

The present study adopts a dialectical position toward ongoing scholarly disputes. It acknowledges the negative influence of traditional-artistic beliefs (Boden, 2004) and the positive role of affective experience (Demmer et al., 2023; Pelowski et al., 2017). Nevertheless, technological comprehension and art-related literacy cannot be interpreted in isolation and must be examined jointly alongside constructs such as acceptance of AI as an artistic tool.

2.3.2 Research Innovations

- **Novel variable**: Incorporating *acceptance of AI as an artistic tool*. Addressing gaps in existing variable sets, this study treats acceptance of AI as a creative artistic instrument as a core predictor and examines how people conceptualize AI's role in art-making.
- **Theoretical integration**: Building a *cognition-emotion-tool-acceptance* theoretical framework. Six dimensions are integrated: traditional-art-view recognition (cognitive dimension), emotional response (affective dimension), AI-technology awareness (technological dimension), AI-tool acceptance (instrumental dimension), art-related background (literacy dimension), and gender (demographic dimension). This multi-factor model synthesizes predictors from separate studies and adds the previously neglected construct of AI-tool acceptance.
- **Improved statistical practice**: Broader sampling covering respondents of varied gender, occupation, and art-experience profiles. Multiple-linear-regression analysis systematically estimates coefficient magnitudes and significance levels for each predictor, unpacking general

mechanisms behind public judgments of AI-art soul and supplying empirical evidence for ongoing theoretical debates.

3. Research Design

3.1 Variable Selection and Theoretical Model

3.1.1 Dependent Variable

The dependent variable is public recognition and perception of whether AI can create soul-endowed artworks, denoted as Y (perception of artistic soul). Y is operationalized via multiple Likert-scale survey items, ranging from 1 to 5; higher values correspond to stronger agreement that AI art can possess a soul.

3.1.2 Independent Variables

Independent Variable	Symbol	Theoretical Rationale
Recognition of traditional artistic views	X_1	Traditional artistic beliefs directly shape judgments of whether soul can exist without human emotion.
Familiarity with AI technologies	X_2	Deeper technical understanding clarifies AI-art-generation logics and may mitigate dismissive biases.
Frequency of emotional responses	X_3	Emotional resonance constitutes a core carrier for soul perception; frequent affective reactions increase likelihood of detecting spiritual meaning in AI art.
Acceptance of AI as an artistic tool	X_4	Framing AI as a tool rather than a human substitute focuses evaluative attention on expressive outputs and may elevate soul-perception levels.
Art-related exposure background	X_5	Art-related experience influences aesthetic judgment; sustained art exposure may facilitate professional-style interpretation of ideational content within works.
Gender	X_6	Dummy variable: male = 1, female = 0

3.1.3 Theoretical Model

$$Y = \beta_0 + \sum_{i=1}^{6} \beta_i X_i + \epsilon$$

- Y = soul-perception dependent variable; $X_1 \sim X_6$ = six independent variables;
- ϵ = error term, $\epsilon \sim N(0, \sigma^2)$

3.2 Data Source and Collection

- Survey participants: respondents with diverse backgrounds
- Survey date: December 7, 2025
- Sample size: N=59 valid responses
- Collection method: online questionnaire
- Sampling: non-probability sampling
- Anonymity: guaranteed
- Questionnaire content: see Appendix

3.3 Variable Measurement

3.3.1 Dependent-Variable Measurement

The dependent variable was initially computed from three survey items:

Item ID	Survey Question	Coding Rule (positive scoring: 1 = negative attitude, 5 = positive attitude)
Q18	Without being told who created a work, can you judge whether it has a "soul" merely from the artwork itself?	Cannot at all = 1; mostly cannot = 2; uncertain = 3; mostly can = 4; fully can = 5
Q19	To what extent do you agree with: "Since AI lacks autonomous consciousness, AI-generated art cannot possibly have a soul"?	Strongly disagree = 5; somewhat disagree = 4; neutral = 3; somewhat agree = 2; strongly agree = 1 (reverse-coded for positive scoring: greater disagreement yields higher soul-perception scores).
Q21	To what extent do you agree with: "AI can expand carriers of artistic soul; human-AI collaboration can generate unique spiritual core meanings"?	Strongly disagree = 1; somewhat disagree = 2; neutral = 3; somewhat agree = 4; strongly agree = 5

Initially the dependent-variable score was the arithmetic average:

$$Y = \frac{1}{3}(Q_{21} + Q_{19} + Q_{18})$$

However, reliability testing revealed low psychometric quality for Q19. Q19 was therefore excluded from dependent-variable calculation. The revised formula is:

$$Y = \frac{1}{2}(Q_{21} + Q_{18})$$

3.3.2 Independent-Variable Measurement

Independent Variable	Corresponding Survey Item	Coding (1-5 positive scoring)
Recognition of traditional artistic views	Q10: Agreement with "The 'soul' of art must originate from human subjective emotion and lived experience."	Strongly disagree = 1; somewhat disagree = 2; neutral = 3; somewhat agree = 4; strongly agree = 5
Familiarity with AI technologies	Q3: Self-reported familiarity with generative-AI techniques such as GAN and Diffusion models.	Completely unfamiliar = 1; slightly familiar (only heard of) = 2; moderately familiar (understand basic functions) = 3; highly familiar (understand underlying principles) = 5
Frequency of emotional responses	Q7: Frequency of affective reactions (e.g., being moved, reflective) upon first viewing AI-art pieces.	Never = 1; rarely = 2; occasionally = 3; frequently = 4; always = 5
Acceptance of AI as an artistic tool	Q12: Willingness to regard AI as an artistic-creation tool analogous to paintbrushes or pigments.	Completely unwilling = 1; somewhat unwilling = 2; uncertain = 3; somewhat willing = 4; fully willing = 5
Art-related exposure background	Q2: History of art-related study or professional practice plus frequency of art-work exposure.	No experience and minimal exposure = 1; no formal training but frequent exposure (exhibitions, collecting) = 2; 1-4 years of experience = 3; 5 years or more = 4
Gender	Dummy variable	Male = 1; female = 0

3.4 Questionnaire Reliability and Validity

3.4.1 Reliability Analysis

A. Questionnaire dimensional structure

Demographic items (Q1 gender, Q2 art background) and complex ranking/multiple-choice items (Q4, Q9) were excluded. Remaining items were grouped into four latent-variable dimensions:

Latent-Variable Dimension	Survey Items	Item Count	Dimension Meaning
Core soul perception (A)	Q17, Q18, Q19, Q21	4 items	Core attitudes and perceptions regarding "soul" in AI art
Technological cognition and attitudes (B)	Q3, Q14, Q16	3 items	Knowledge of AI technologies plus attitudes toward AI-art characteristics
Traditional artistic views (C)	Q10, Q5	2 items	Conventional beliefs concerning the origins of artistic "soul"
Affective and behavioral tendencies (D)	Q7, Q8, Q12, Q13	4 items	Affective responses to AI art plus behavioral dispositions

B. Cronbach's α reliability formula

$$\alpha = \frac{k}{k-1}(1 - \frac{\sum_{i=1}^{k} S_i^2}{S_{total}^2})$$

Where k = number of items; S_i^2 = sample variance of item i; S_{total}^2 = sample variance of the summed total score across k items.

Evaluation criteria: overall α≥0.7 = acceptable; α≥0.8 = good, α≥0.65 is treated as acceptable.

C. Reliability test results

Dimension	Cronbach's α	Reliability Evaluation
Core soul perception (A)	−0.435	Poor; unacceptable
Technological cognition and attitudes (B)	0.654	Marginally acceptable
Traditional artistic views (C)	0.818	Good
Affective and behavioral tendencies (D)	0.831	Good
Full questionnaire	0.856	Good

D. Result interpretation

The extremely low α for dimension A likely stemmed from respondents misinterpreting the reverse-worded Q19 item. After removing Q19:

Dimension	Cronbach's α	Reliability Evaluation
Core soul perception (A) (Q19 removed)	0.766	Good
Technological cognition and attitudes (B)	0.654	Marginally acceptable
Traditional artistic views (C)	0.818	Good
Affective and behavioral tendencies (D)	0.831	Good
Full questionnaire	0.856	Good

Exclusion of Q19 substantially improved reliability. Q19 was consequently dropped from dependent-variable computation.

3.4.2 Validity Analysis

A. Content validity

Validation Aspect	Verification Process	Outcome
Topic relevance	1. Core-soul-perception items directly address perceptions, controversies, and potential of soul in AI art. 2. Technological-cognition items focus on AI-art knowledge and characteristics, with indirect bearing on soul perception. 3. Traditional-art-view items target core disputes over where artistic soul originates. 4. Affective-behavior-tendency items measure emotional responses and tool-related attitudes influencing soul-perception judgments. Conclusion: all 15 retained items relate directly or indirectly to the central theme.	Strong relevance; no irrelevant items
Content comprehensiveness	1. Four key dimensions covered: core perception, technological cognition, traditional beliefs, affective-behavioral responses. 2. Each dimension uses symmetric 5-point Likert scales. 3. Logical sequence: cognition → attitude → perception → behavior is fully represented. Conclusion: no obvious omissions.	Comprehensiveness criterion satisfied
Appropriateness of wording	1. Clear item phrasing. 2. Symmetric 1-5 response-option gradients. 3. Accessible wording for participants with varying art-experience levels. Conclusion: minimal comprehension barriers.	Wording and response options are scientifically designed with negligible measurement bias

B. Construct validity: KMO and Bartlett's sphericity test

- Judgment benchmarks: KMO > 0.9 = excellent; 0.8 = good; 0.7 = mediocre; 0.6 = poor; < 0.5 = unacceptable for factor analysis.
- Test output:

KMO = 0.909

Bartlett sphericity statistic = 406.561

Bartlett sphericity p<0.001

- Interpretation: KMO=0.909≥0.7; the null hypothesis that the correlation matrix equals an identity matrix is rejected. The dataset is suitable for exploratory-factor analysis.

3.5 Statistical-Analysis Methods

3.5.1 Regression Analysis

A. Ordinary-Least-Squares (OLS) Regression

OLS minimizes the sum of squared residuals and yields Best Linear Unbiased Estimates (BLUE) under the following classical assumptions:

1. Linear relationships between dependent and independent variables.
2. Random sampling yielding representative observations.
3. Absence of perfect multicollinearity among predictors.
4. Zero conditional mean: expected value of residuals equals zero; residuals are uncorrelated with predictors.
5. Homoscedasticity: constant residual variance (no heteroskedasticity).
6. Residuals follow a normal distribution (supporting t-tests and F-tests for statistical inference).

B. Model-evaluation indicators

1. **F-statistic**: tests whether the full set of predictors jointly explains variance in the dependent variable. Larger F-values indicate stronger collective explanatory power, interpreted alongside Prob>F.
    - Null hypothesis: all regression coefficients = 0 (the model has no explanatory capacity).
    - Decision rule: p<0.05 rejects the null hypothesis; the overall model is statistically significant.
2. **t-statistic**: evaluates whether an individual predictor's regression coefficient differs statistically from zero. Larger absolute t-values signal stronger individual predictive influence, interpreted against P>|t|.
    - Null hypothesis: the given predictor's coefficient = 0 (no effect).
    - Decision rule: p<0.05 rejects the null hypothesis for that predictor.
3. **R^2 (coefficient of determination)**: proportion of total dependent-variable variance explained

by the model, ranging from 0 to 1. Higher values indicate better fit. In social-science research, $R^2 > 0.3$ is conventionally treated as acceptable. Limitation: R^2 never decreases when adding new predictors, regardless of whether those predictors meaningfully improve model performance.

4. **Adjusted- R^2**: penalizes R^2 for the number of predictors. Adjusted- R^2 rises only when an added variable meaningfully boosts explanatory power; it provides a more trustworthy fit metric than raw R^2.

3.5.2 Regression Diagnostics

A. Multicollinearity: Variance-Inflation Factor (VIF)

$$VIF_i = \frac{1}{1 - R_i^2}$$

Where R_i^2 is the R^2 obtained by regressing predictor i on all remaining independent variables.

Interpretation benchmarks:

- VIF < 5: no substantial multicollinearity
- 5 ≤ VIF < 10: moderate multicollinearity
- VIF ≥ 10: severe multicollinearity

B. Residual normality

Two graphical diagnostics are deployed:

1. Residual histogram plus kernel-density curve: residuals should approximate a bell-shaped normal distribution, with the kernel-density curve closely overlapping the theoretical-normal-density trace.
2. Residual Q-Q plot: observed residual quantiles should align approximately along a 45-degree reference line derived from theoretical normal quantiles. Marked departures indicate non-normality (skewness, heavy tails, etc.).

C. Heteroskedasticity

Heteroskedasticity arises when residual variance varies systematically across predictor ranges. OLS homoscedasticity assumptions are violated, coefficient-estimate variances become inflated, and statistical inference becomes unreliable (type-I-error risk increases). A fitted-values-versus-residuals scatterplot visually assesses heteroskedasticity: residuals should distribute randomly around zero without visible systematic patterns.

4. Main Empirical Findings

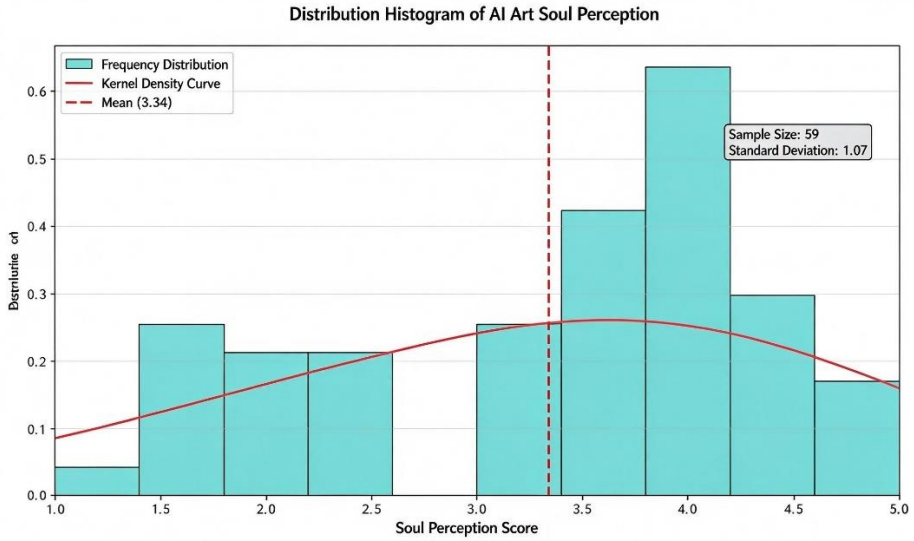
4.1 Core Findings

- Reliability testing demonstrated that excluding Q19 substantially improves model explanatory power; adjusted-R^2 rises from 0.20 to 0.513.
- The overall regression model yields $F=11.18$, $p<0.001$, with predictors collectively explaining 51.3 % of variance in soul-perception scores (moderate-to-substantial explanatory performance).
- Only two predictors attain statistical significance: recognition of traditional artistic views and gender.

4.2 Descriptive Statistics for Key Variables

4.2.1 Dependent-variable descriptive results

- Mean soul-perception score = 3.34 (5-point scale); standard deviation = 1.07. Values cluster within the 3.8-4.2 range.
- Perception-level breakdown:
  - Lean toward judging AI art as soulless (1-3 points): 28.8 %
  - Neutral stance (3 points): 10.2 %
  - Lean toward judging AI art as potentially soul-bearing (3-5 points): 61.0 %

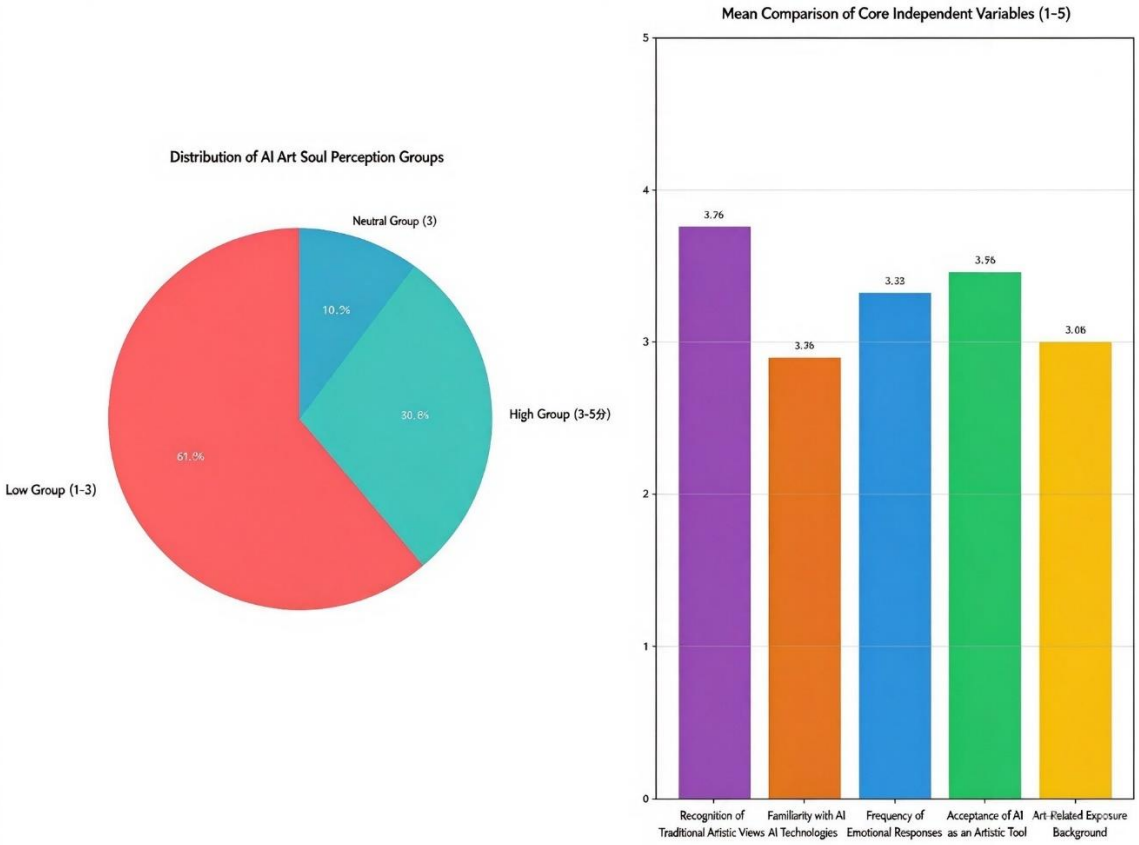


4.2.2 Descriptive profiles for independent variables

1. **Recognition of traditional artistic views**: relatively high (mean = 3.76). 71.2 % of respondents somewhat or strongly agreed that artistic soul must derive from human emotion and lived experience.

2. **Familiarity with AI technologies**: moderately low (mean = 2.90). Only 22.0 % reported high technical understanding; 40.7 % were slightly familiar or completely unfamiliar.
3. **Frequency of emotional responses**: moderately positive (mean = 3.32). 47.5 % frequently or always experienced affective reactions to AI-art pieces.
4. **Acceptance of AI as an artistic tool**: moderately positive (mean = 3.46). 59.3 % were somewhat or fully willing to conceptualize AI as an artistic-creation tool comparable to traditional media.
5. **Art-related exposure background**: many respondents lacked formal art experience. 40.7 % had minimal art-related exposure; 32.2 % had 1-4 years of art-related experience; 27.8 % had five or more years of experience.
6. **Gender distribution**: male 52.5 %, female 47.5 %; well-balanced sample gender composition.

4.2.3 Descriptive Charts of Core Variables



- Upper left plot (pie chart): Graded distribution of perceived soulfulness of AI art, which intuitively presents the proportional structure of “tending to perceive no soul”, “neutral”, and “tending to perceive a soul”.
- Upper right plot (bar chart): Mean comparison of five independent variables excluding gender, clearly showing that the recognition of traditional art concepts has the highest mean value while the familiarity with AI technology has the lowest.

Cross-tabulation: traditional-art-view recognition × soul-perception level

Soul-Perception Level Recognition of Traditional Artistic Views	Lean toward soulless (1-3)	Neutral (3)	Lean toward soul-bearing (3-5)	Total	Percentage
Strongly disagree	7	1	0	8	13.56%
Somewhat disagree	3	0	0	3	5.08%
Neutral	2	3	1	6	10.17%
Somewhat agree	2	0	18	20	33.90%
Strongly agree	3	2	17	22	37.29%
Total	17	6	36	59	100.00%
Row percentage	28.81%	10.17%	61.02%	100.00%	

4.3 Regression-analysis Results

4.3.1 Regression results

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        soul_perception   R-squared:                       0.563
Model:                            OLS   Adj. R-squared:                  0.513
Method:                 Least Squares   F-statistic:                     11.18
Date:                Sat, 27 Dec 2025   Prob (F-statistic):           5.57e-08
Time:                        00:39:22   Log-Likelihood:                -62.674
No. Observations:                  59   AIC:                             139.3
Df Residuals:                      52   BIC:                             153.9
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const              0.7499      0.363      2.066      0.044       0.022       1.478
traditional_art    0.3475      0.101      3.440      0.001       0.145       0.550
ai_knowledge       0.1103      0.091      1.214      0.230      -0.072       0.293
emotion_response   0.1360      0.093      1.468      0.148      -0.050       0.322
ai_tool_identity   0.1078      0.113      0.950      0.347      -0.120       0.335
art_background    -0.0406      0.080     -0.506      0.615      -0.202       0.121
gender             0.5470      0.214      2.559      0.013       0.118       0.976
==============================================================================
Omnibus:                        1.069   Durbin-Watson:                   1.674
Prob(Omnibus):                  0.586   Jarque-Bera (JB):                0.848
Skew:                          -0.292   Prob(JB):                        0.655
Kurtosis:                       2.940   Cond. No.                         29.2
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4.3.2 Regression-coefficient interpretation

Independent Variable	Coefficient β	Standard Error	t-value	p-value	Substantive Interpretation (per one-unit score increase)
Intercept	0.7499	0.363	2.066	0.044	—
Frequency of emotional responses	0.1360	0.093	1.468	0.148	Statistically non-significant
Recognition of traditional artistic views	0.3475	0.101	3.440	0.001	Soul-perception score rises on average by 0.3475 points; positive and statistically significant
Familiarity with AI technologies	0.1103	0.091	1.214	0.230	Statistically non-significant
Acceptance of AI as an artistic tool	0.1078	0.113	0.950	0.347	Statistically non-significant
Art-related exposure background	−0.0406	0.080	−0.506	0.615	Statistically non-significant
Gender (male = 1)	0.5470	0.214	2.559	0.013	Male respondents average 0.547-point higher soul-perception scores; positive and statistically significant

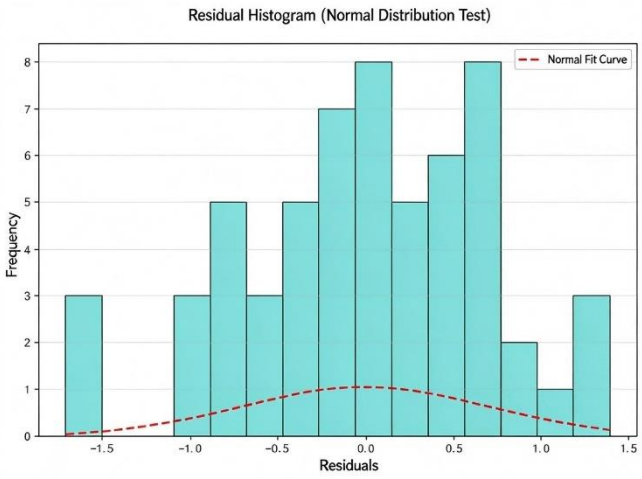
4.4 Regression Diagnostics

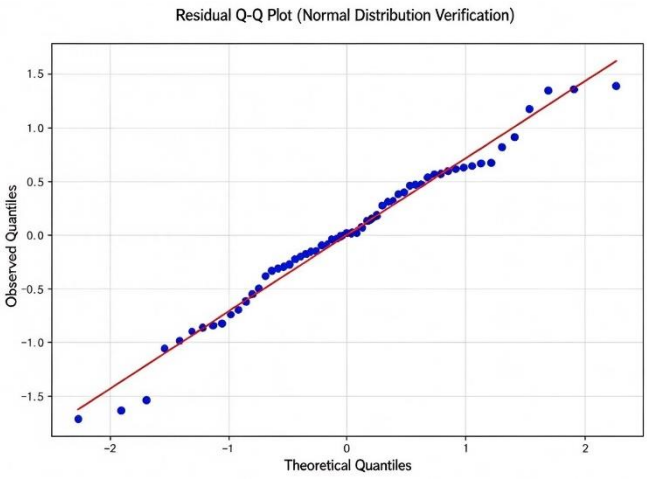
4.4.1 Multicollinearity (VIF)

Predictor	VIF Value	Multicollinearity Assessment
Intercept	13.974	High VIF for intercept is a common artifact and requires no remedial action
Recognition of traditional artistic views	1.994	No meaningful multicollinearity
Familiarity with AI technologies	1.475	No meaningful multicollinearity
Frequency of emotional responses	1.373	No meaningful multicollinearity
Acceptance of AI as an artistic tool	2.052	No meaningful multicollinearity
Art-related exposure background	1.302	No meaningful multicollinearity
Gender	1.209	No meaningful multicollinearity

All substantive predictors yield VIF values well below 5; multicollinearity poses no threat to inference. No variable-removal, merging, or centering procedures are required.

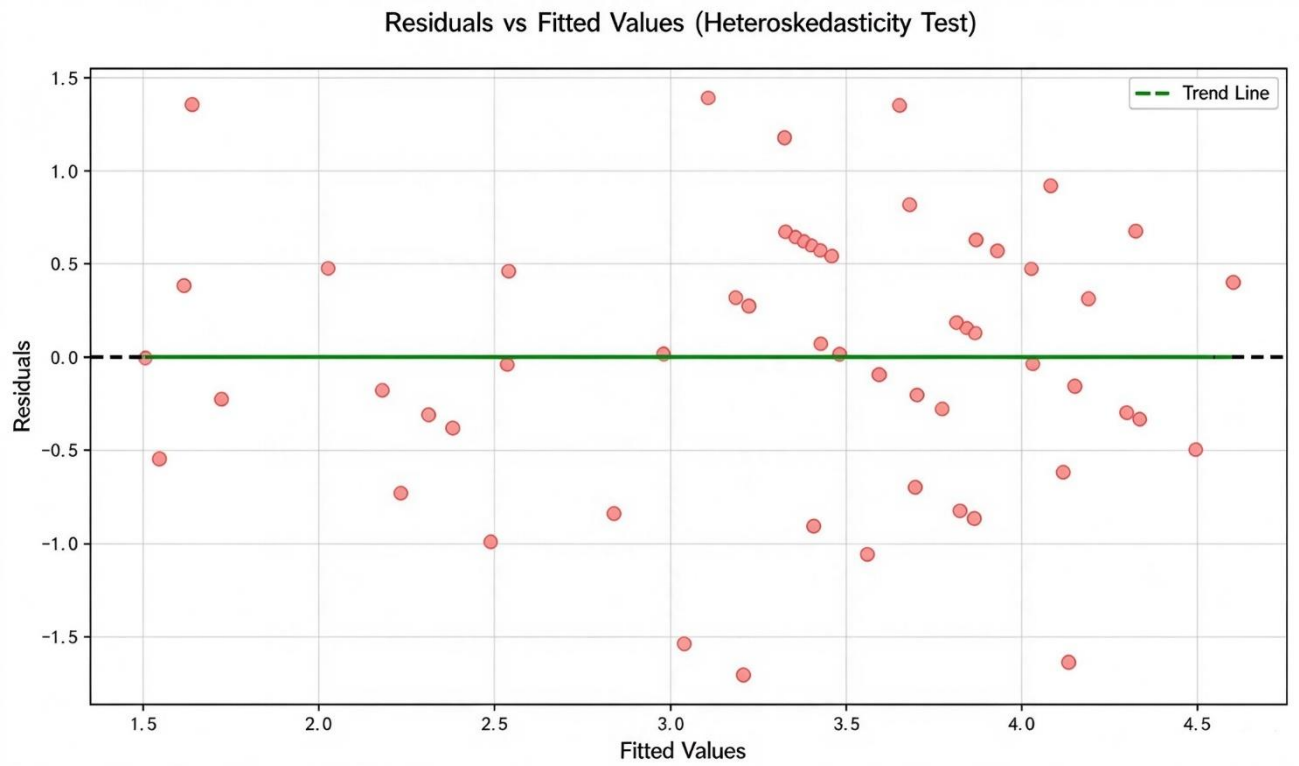
4.4.2 Residual normality





- Residual histograms and Q-Q plots show residuals approximately conform to normal-distribution assumptions.
- Data points in the Q-Q diagram cluster close to the reference straight line.

4.4.3 Heteroskedasticity diagnostics



In residuals-versus-fitted-values scatterplots, residual points distribute randomly above and below the zero horizontal reference line without systematic patterning. Homoscedasticity assumptions are satisfied; heteroskedasticity is absent.

5. Conclusions and Recommendations

5.1 Research Conclusions

5.1.1 Main Conclusions

Final regression-model equation:

$$Y = 0.7499 + 0.3475X_1 + 0.1103X_2 + 0.1360X_3 + 0.1078X_4 - 0.0406X_5 + 0.5470X_6 + \epsilon$$

1. **Status of public soul-perception toward AI art**: overall positive orientation. Mean soul-perception score = 3.27 out of 5. 28.8 % of respondents tended to judge AI art as soulless; 61.0 % leaned toward attributing soul-like qualities to AI art; 10.2 % remained neutral. Most participants hold optimistic attitudes regarding the potential spiritual content of AI-generated art.

2. **Significant predictors**: recognition of traditional artistic views and gender exert statistically significant influences ($p<0.05$). Familiarity with AI technologies, art-related background,

frequency of emotional responses, and acceptance of AI as an artistic tool show no statistically significant effects ($p>0.05$).

- Recognition of traditional artistic views: significant positive coefficient. Respondents who more strongly endorse the proposition that artistic soul must stem from human emotion and lived experience tend to assign higher soul-perception scores to AI art. Traditional artistic cognitions positively predict receptiveness toward soul-bearing possibilities within AI-generated works.
- Gender: significant positive coefficient for male gender. Male participants are more inclined to believe AI can produce soul-infused artistic outputs.

3. **Model performance**: overall regression model is statistically significant; adjusted-$R^2 = 0.513$. The selected predictors jointly explain 51.3 % of total variance in soul-perception scores, demonstrating good explanatory power and capturing core logics governing public judgments of AI-art soul.

5.1.2 Secondary Conclusions

1. **Global questionnaire reliability and validity are satisfactory.** Item Q19, a reverse-worded question, distorted dimension-level reliability; its exclusion markedly improved regression-model explanatory capacity. Survey designers should minimize reverse-worded items to reduce respondent misreading risk.
2. **Public self-reported familiarity with AI-generation technologies is generally low (mean = 2.90).** Nevertheless, technical knowledge exhibits no significant predictive effect on soul perception. Public judgments about soul in AI art appear to draw more heavily on subjective aesthetic and artistic beliefs rather than technical comprehension of algorithmic mechanics.
3. **Art-related background (formal training or professional experience) yields no significant coefficient**. Lay audiences and art-experienced participants do not differ meaningfully in soul-perception judgments. Evaluations of soul-like qualities in AI art display substantial populist universality.

5.2 Practical Recommendations

Findings inform strategies for public-facing outreach, platform-creator practice, and scholarly-industry development.

A. For the general public: bridge traditional-artistic heritage and reduce comprehension barriers

1. Build conceptual bridges connecting traditional art and AI-generated art. Deploy exhibitions,

public lectures, and popular-science content illustrating how AI art inherits and evolves traditional aesthetic and spiritual values. Help audiences recognize that traditional-artistic worldviews can support acceptance of soul-like potential in AI-artistic outputs.

2. De-emphasize complex technical details. Since technical familiarity and emotional-response frequency are non-significant predictors of soul perception, public communication should prioritize artwork-intrinsic merit instead of algorithmic principles. Lower barriers for direct aesthetic engagement with AI-art pieces.

B. For media platforms and artistic creators: attend to gender-related attitudinal differences to amplify perceived soul-like qualities

1. Gender-targeted communication strategies. Female audiences may respond more strongly to affect-driven, narratively rich content. Create AI-art pieces emphasizing humanistic storytelling while minimizing overly technical framing to foster greater affective resonance among women.

2. Embed core traditional-artistic spirits within AI-driven creation. Integrate themes and aesthetic elements from traditional art into AI-generated outputs to align works with public traditional-artistic intuitions and enhance subjective perceptions of "soul".

C. For industry and academia: advance ecosystem construction and deepen research

1. Differentiated promotion strategies by gender. Highlight technological innovation and artistic breakthrough potential for male-oriented outreach; foreground humanistic value and emotional meaning for female-oriented communication. Balanced differentiated messaging can broaden societal acceptance of AI art.

2. Further research on interplay between traditional-artistic beliefs and AI-art reception. Expand sampling frameworks, enrich predictor sets, and refine theoretical models explaining public soul-perception formation.

3. Investigate non-significant predictors in greater depth. Explore contextual or moderating factors that suppress observable effects for variables such as AI-technology familiarity and emotional-response frequency, supplying more comprehensive evidence for public-opinion-guidance practice.

5.3 Research Limitations and Future-research Outlook

5.3.1 Limitations

1. Limited sample size (N=59) and coverage. The non-probability sample lacks sufficient stratification across age brackets, occupations, and geographic groups, potentially constraining generalizability.

2. Incomplete variable scope. This study omits potential predictors such as genre or stylistic

categories of AI-artworks; further variance remains to be explained.

3. Restricted methodological toolkit. The investigation relies exclusively on quantitative regression analysis. Controlled experimental designs (e.g., between-subject stimulus-presentation conditions) could isolate viewers' capacity to subjectively detect soul-like attributes in artworks.

4. Questionnaire-item-wording risk. Reverse-worded items increase misinterpretation probability among respondents. Future surveys should prioritize uniformly positive-direction question phrasing.

5.3.2 Future-research Directions

1. Expand sample scale and population coverage to improve external validity.

2. Adopt mixed-methods research combining quantitative statistics and qualitative interviews; incorporate additional contextual predictors to unpack causal mechanisms behind soul-perception judgments.

3. Track longitudinal shifts in public attitudes alongside AI-technology iteration. Investigate how algorithmic evolution reshapes long-run public perceptions of soul-bearing potential within AI-generated art, informing sustainable integration between AI-art innovation and traditional-artistic heritage.

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Appendix

Appendix A: Full Questionnaire

1. What is your gender? □ Male □ Female
2. Do you have art-related study or professional experience (e.g., painting, design, art-history research)?

   □ No experience, and rarely encounter artworks

   □ No formal training but frequently encounter artworks (exhibitions, collecting etc.)

   □ Yes (1-4 years)

   □ Yes (5 years or longer)
3. How familiar are you with generative-AI techniques such as GAN and Diffusion models?
   □ Completely unfamiliar

   □ Slightly familiar (only have heard of AI image generation)

   □ Moderately familiar (understand core functions and application contexts)

   □ Highly familiar (can explain underlying technical principles)
4. Among the three AI-art-generation modes below, which do you think produces works most likely to convey spiritual core content (emotion, ideas)? Please rank them from highest to lowest importance.

   □ AI-assisted creation (humans perform core creation; AI handles color-filling, detail-optimization and other technical support)

□ Human-AI co-creation (humans set themes/styles; AI generates drafts, followed by human revision and feedback loops)

□ Pure-AI generation (no human intervention; algorithms autonomously generate outputs from learned datasets)

Rank:_____

5. For a human-AI-co-creation workflow: *human defines environmental-protection theme → AI generates draft → human selects and revises*. Where do you believe the work's creative intention (intended ideas) primarily originates?

   □ Entirely from humans

   □ Mainly from humans; AI supplies minor auxiliary contributions

   □ Equally from humans and AI

   □ Mainly from AI; humans make trivial adjustments

   □ Entirely from AI

6. To what extent do you agree with: "AI-assisted creations carry more 'human warmth' than purely AI-generated works"?

   □ Strongly disagree □ Somewhat disagree □ Neutral □ Somewhat agree □ Strongly agree

7. When you first view an AI-generated visual-art piece (e.g., digital painting), do you experience distinct emotional responses (being moved, reflective, astonished, etc.)?
   □ Never □ Rarely □ Occasionally □ Frequently □ Always

8. After encountering an AI-artwork, do you actively revisit its content or attempt to interpret its intended meaning within the following 1-2 days?

   □ Never □ Rarely □ Occasionally □ Frequently □ Always

9. If an AI-artwork evokes strong emotional resonance in you, what do you believe this resonance primarily derives from? (Multiple selections allowed)

   □ Visual formal qualities (color palette, composition)

   □ Ideas or themes expressed by the work

   □ Back-story of the creation process

   □ Other (please specify):_____

10. To what extent do you agree with the traditional claim: "The 'soul' of art must originate from

human subjective emotion and lived experience”?

□ Strongly disagree □ Somewhat disagree □ Neutral □ Somewhat agree □ Strongly agree

11. In your judgment, what are the most central criteria for deciding whether an artwork possesses a “soul”? (Multiple selections allowed)

    □ Whether the creator possesses autonomous consciousness

    □ Whether the artwork can evoke viewers’ emotional resonance

    □ Whether the artwork conveys unique thoughts or meanings

    □ Whether the artwork possesses unrepeatable uniqueness

    □ Other (please specify):_____

12. How willing are you to treat AI as an artistic-creation tool comparable to paintbrushes and pigments?

    □ Completely unwilling □ Somewhat unwilling □ Uncertain □ Somewhat willing □ Fully willing

13. After discovering that a highly resonant “excellent artwork” you appreciated is AI-generated, your evaluation of this work would:

    A. Improve greatly B. Improve slightly C. Remain unchanged D. Deteriorate slightly E. Deteriorate greatly

14. Do you proactively follow developments in AI-art creation (new artworks, new technologies)?
    □ Never □ Rarely □ Occasionally □ Frequently □ Always

15. Which features of AI-artworks most strongly give you the impression of a “soul”? (Multiple selections allowed; rank from highest to lowest importance)

    □ Subtlety of emotional expression (capacity to convey complex sentiments)

    □ Formal innovativeness (breaking conventional visual-art styles)

    □ Degree of cultural-element integration (national-cultural symbols, historical motifs)

    □ Depth of thematic content (exploring environmentalism, human nature, and other profound topics)

    Rank:_____

16. How large an influence does “subtlety of emotional expression” exert on whether you perceive an AI-artwork as having a soul?

□ No influence □ Minor influence □ Moderate influence □ Substantial influence □ Extreme influence

17. If you are informed of creation context (human-defined themes, revision workflows) before viewing an AI-artwork, how does this alter your perception of its “soul”? □ Diminish greatly □ Diminish slightly □ No change □ Enhance slightly □ Enhance greatly

18. Without prior notification whether a piece is AI-produced or human-made, can you judge whether it has a “soul” solely from the artwork itself?

    □ Cannot at all □ Mostly cannot □ Uncertain □ Mostly can □ Fully can

19. Claim: “As a tool without autonomous consciousness incapable of transmitting lived human experience, AI-generated art cannot possess a soul.” To what extent do you agree? □ Strongly agree □ Somewhat agree □ Neutral □ Somewhat disagree □ Strongly disagree

20. Claim: “Mass-batch AI-art generation causes homogenization and dilutes the unique ‘soul’ of artworks.” To what extent do you agree?

    □ Strongly agree □ Somewhat agree □ Neutral □ Somewhat disagree □ Strongly disagree

21. To what extent do you agree with: “AI expands potential carriers for artistic ‘soul’; for instance, human-AI collaboration may generate spiritual-core content unachievable by unaided human creation (e.g., cross-cultural-fusion themes)”?

    □ Strongly disagree □ Somewhat disagree □ Neutral □ Somewhat agree □ Strongly agree

22. After sustained exposure to AI art for more than one year, how would your personal capacity to perceive “soul” within AI-artworks change?

    □ Deteriorate greatly □ Deteriorate slightly □ Remain unchanged □ Improve slightly □ Improve greatly

Appendix B: Python-Analysis Code

(The full Python code segment is omitted here to conserve space; the original Chinese-version code contains complete routines for reliability-validity testing, descriptive statistics, OLS-regression fitting, multicollinearity diagnosis, residual-normality and heteroskedasticity diagnostics, plus visualization workflows.)